

KUNLE KAZEEM ADEYEMO

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EDUCATION

Michigan State University (MSU), MI, USA. PhD in Mechanical Engineering (<i>Solid Mechanics, Design, and Manufacturing</i>) – in view PhD in Civil Engineering (<i>Structural Engineering, Mechanics and Materials</i>) – in view	Aug 2022 – Present GPA: 3.87/4.00
Michigan State University (MSU), MI, USA. Master of Science in Mechanical Engineering	Jan 2024 – Dec 2025 GPA: 3.90/4.00
Warsaw University of Technology, WA, Poland. Master of Science in Power Engineering	Feb 2021 – Jun 2022 GPA: 3.82/4.00
Ladoke Akintola University of Technology, Oyo, Nigeria. Bachelor of Technology in Mechanical Engineering	Nov 2012 – Sept 2022 GPA: 3.72/4.00

RESEARCH & WORK EXPERIENCE

Composite Vehicle Research Center - MSU <i>Research Assistant III- Computational modeling and Materials testing.</i> • Developing experimentally validated models for progressive failure analysis in a hybrid fastening system. • Proposed a novel confinement implicit model for bolt-hole clearance filled with adhesive using Abaqus FEA.	Aug 2023 – Present
Civil and Environmental Engineering - MSU <i>Teaching Assistant II - Introduction to Structural Analysis course (CE 305)</i> • Assist in solving problems during recitation sessions. • Practical training in SAP 2000 modeling software for structural analysis.	Jan 2024 – Present
Sustainable Energy Laboratory - MSU <i>Research Assistant I</i> • Prototyped and proposed new degradation testing protocols for EV lithium-ion battery systems to enhance safety and performance assessment. • Managed experimental planning and documentation under strict laboratory safety protocols.	Aug 2022 – Jun 2023
Amazon-Warsaw, Poland <i>Senior Compliance Associate</i> • Reviewed product information against regulatory documentation to ensure compliance for Amazon listings. • Products classification, identification, and labeling for improvement opportunities.	May 2022 – July 2022

LEADERSHIP & OUTREACH ACTIVITIES

Society for the Advancement of Material and Process Engineering (SAMPE): MSU Chapter <i>Treasurer</i> • Revitalization of the SAMPE-MSU Student Chapter. • Submitted a grant proposal for the SAMPE 1000 USD grant application.	Sept 2025 – Present
University Committee on the Libraries: MSU <i>Graduate Student Representative</i> • Participating in policymaking of the MSU libraries • Monitoring graduate students' concerns on libraries' resources, policies, and services.	Sept 2025 – Present
Graduate Studies Committee: Department of Mechanical Engineering MSU <i>Student Representative</i> • Refined the Mechanical Engineering handbook for graduate students	Sept 2025 – Present

IACMI- The Composite Institute: Winona State University**August 2025***Summer Fellow*

- Hands-on experience on injection molding for manufacturing coupons with thermoplastic materials (PET, PP)
- Use of thermoforming to manufacture a wind shield with thermoplastics.
- Compression molding for producing a plate of glass fiber-reinforced thermoplastic composite.

Mechanical Engineering Graduate Student Association (ME-GSA): MSU**July 2025 – Present***Council of Graduate Student Representative*

- Orientation for the incoming graduate students in Mechanical Engineering
- Presented the COGS overview during the orientation

ICMED- Integrated Computational Material Education: University of Michigan**July 2025***Summer Fellow*

- Engaged in a computational materials science and engineering crash course.
- Hands-on experience on computational mechanics and density functional theory.

Summer Research Opportunity Program (SROP): MSU**May 2025 – July 2025***Assistant Graduate Facilitator*

- Facilitated the weekly research presentation program for the SROP undergraduate students in STEM.
- Provided critical and constructive feedback for the students.
- Review students' posters and oral presentations.
- Served as poster and oral presentation judge.

Society of Plastics Engineers: MSU**Sept 2024 – Present***Technical Project Lead*

- Engagement in a series of workshops for STEM undergraduates in MSU.
- Conducted a lightweight plastics design bridge competition for undergraduates.
- Collaborated with AMSE-MSU for Spring'25 Water-wheel plastics design.

ComSciCon-MI: Michigan**Aug 2024 – Nov. 2025***Application Review Manager*

- Set clear guidelines and criteria for evaluating applicants based on the workshop's goals and objectives.
- Assigned applications to reviewers and provided necessary instructions or support to them.

Institute of Cyber-Enabled Research**Sept 2024 – Dec 2024***Cloud Computing Fellow*

- Training on AWS Academy Cloud Foundations.
- Channeling the advantages and limitations of commercial cloud computing regarding research yield and adoption.

Immersive Visualization Institute: MSU**June 2024***Summer Fellow*

- Participated in using the three large-scale digital visualizations
- Presented IVI proposal using QGIS on "the trend of electric vehicle usage in the US."

PUBLICATIONS**Journal Publications (Published/ Communicated/ In Review/ In progress)**

- 1) Mahmoodul Haq, **Kunle, K. Adeyemo**, Syed F. Hassan, Cloud, G.L. "Novel Hybrid Fastening Systems for Composite Joints: A Parametric Study - Part I." Journal of Composite Structures – [In review.](#)
- 2) **Kunle, K. Adeyemo**, Andrew Valentine, Syed Fahad Hassan, Mahmoodul Haq. Progressive Failure Analysis for GFRP/AL Single-Lap Bolted Joints. Mechanics of Advanced Materials and Structures – [In progress.](#)
- 3) Sampson E... **Kunle, K. Adeyemo** (2026). "Hybrid Renewable Energy Optimization Using HOMER Pro: A Systematic Literature Review and Future Research Agenda for Emerging Economies. "IEEE ACCESS – DOI: <https://doi.org/10.1109/ACCESS.2026.3654991>

Conference Proceedings

- 1) **Kunle, K. Adeyemo**, Salina Ramli, Syed Fahad Hassan, and Mahmoodul Haq (2026). Hybrid Fastening System with Composite Substrates: An Experimental and Numerical Approach. **Published** – [ANTEC 2026](#).
- 2) **Kunle, K. Adeyemo**, Valentine Andrew, Syed Fahad Hassan and Mahmoodul Haq (2026). Hybrid Fastening System with Dissimilar Substrates: An Experimental and Numerical Approach. **Communicated** – [SAMPE 2026](#)
- 3) **Kunle, K. Adeyemo**, Valentine Andrew, and Mahmoodul Haq (2025). Progressive Failure Analysis of Dissimilar Single-Lap Bolted Joint Using LaRC05 Failure Criteria. **Published** – [ACCE 2025](#).
- 4) **Kunle, K. Adeyemo**, et. al. (2020). The effect of residence time and temperature on product yields during the pyrolysis of coconut husks. Conference Series 1734 012036. DOI: <https://doi.org/10.1088/1742-6596/1734/1/012036>
- 5) Kolawole, W. Okunoye, O P Ogundile, **Kunle Adeyemo**, et.al (2020). Evaluation of the effect of silver nanoparticles and biopolymer on modified Nigerian bentonite rheological properties. DOI: <https://doi.org/10.1088/1742-6596/1734/1/012042>

ORAL PRESENTATIONS

- 2026 AGSA Oral Conference Presentation
- 2026 ASME Graduate Student Video competition
- 2026 SAMPE JEC Two Minute Innovative Challenge
- 2026 COGS Oral Conference Presentation
- 2026 & 2025 COGS Conference Three Minute Thesis Talk
- 2025 IFORE Student Research Competition
- 2025 AGEF Student Success Conference
- 2025 & 2024 ACCE Five Minutes Poster Talk
- 2024 COMSCICON-MI One Minute Research Talk

POSTER PRESENTATIONS

- **Kunle, K. Adeyemo**, and Mahmood Haq. *Hybrid Fastening for Dissimilar Substrates: A lightweighting Solution for Electric Vehicles. AGSA 2026*
- **Kunle, K. Adeyemo**, and Mahmood Haq. *Progressive Failure Analysis for Hybrid Fastening System with Dissimilar Substrates. [ANTEC 2026](#)*.
- **Kunle, K. Adeyemo**, and Mahmood Haq. *Experimentally Validated Numerical Analysis for Hybrid Fastening System with Dissimilar Substrates. [ACCE 2025](#)*.
- **Kunle, K. Adeyemo**, and Mahmood Haq. *Progressive damage analysis in a Single-Lap Bolted Joint. [ICMED 2025](#)*.
- **Kunle, Adeyemo**, & Mahmood Haq. *Bearing Stress Analysis in a single-lap hybrid fastening system. [MSU-EGR 2025](#)*.
- Abdifitah Adan, **Kunle, Adeyemo**, Hassan Fahad and Mahmood Haq. *Effect of Machined and Printed Holes on 3D printed Continuous Fiber Reinforced Thermoplastics (CFRTPs). [MSU-EGR Symposium 2025](#)*.
- **Kunle Adeyemo**, Abdifitah Adan, Hassan Fahad and Mahmood Haq. *3D printing and investigating of bearing performance of an Open Hole CFRTP. [MSU-COGs 2025](#)*.
- **Kunle Adeyemo**, and Mahmood Haq. *Experimental and Multi-scale Analysis of Progressive Failure in Multi-material Single-lap Bolted Joint. ACCE 2024*.
- **Kunle Adeyemo**, and Mahmood Haq. *Numerical Analysis of Preload Monitoring in Composite Bolted Joint Subjected to Static Loading. ACCE 2024*.
- **Kunle Adeyemo**, and Mahmood Haq. *Experimental and Numerical Analysis of Progressive Failure in Multi-Material Single Lap Bolted Joints - An Overview. AUTO EPCON 2024*.
- **Kunle Adeyemo**, Mahmood Haq. *In service preload monitoring in composite bolted joint subjected to static loading with numerical approach. AUTO EPCON 2024*.
- **Kunle Adeyemo**, and Annick Anctil. *Estimating the economic benefit of using second life batteries in Fast Charging Stations in Michigan. MSU-EGR Symposium 2023*.

TECHNICAL SKILL SET

Simulation & Modeling Tools:	Abaqus, Ansys-Fluid, SOLIDWORKS, AutoCAD, CFD
Programming/ Software:	Python, MATLAB, MS Office (Excel Macro/Data Analysis)
Infrastructure:	AWS Cloud Computing (Foundational)
Testing & Validation:	Tensile (MTS), Impact (Instron), thermal (TGA), Nano (SEM)

AWARDS/FELLOWSHIPS

- 2026 NSF-MSEC Travel Award
- 2026 ASME Graduate Student Video Competition, 3rd place (\$1500)
- 2026 AGSA Poster Award, 1st position (\$300)
- 2026 SAMPE Student Leadership Award
- 2026 SPE Detroit Travel Grant Award (\$1500)
- 2026 SAMPE University Research Symposium, Semi-finalist.
- 2026 ANTEC Conference Poster Travel Award (\$300)
- 2025 Council of Graduate Students Travel Grant (\$300)
- 2025 Best Oral Presentation AGEF, People's Choice Award
- 2025 Best Oral Presentation AGEF, 1st Runner-up (\$250)
- 2025 MSU Research Enhancement Award (\$1800)
- 2025 SPE Detroit-Section Scholarship (\$2100)
- 2025 SPE ACCE Graduate Scholarship (\$2000)
- 2025 Summer fellowship for Integrated Computational Material Education (\$450)
- 2024 SPE Detroit-Section Scholarship (\$2100)
- 2024 Summer fellowship for Immerse Visualization Institute (\$500)
- 2024 Auto EPCON Best Poster Award, 1st Runner-up (\$200)
- 2022 Warsaw University of Technology Chancellor's Award, Best Student
- 2021 Warsaw University of Technology Chancellor's Award, Best Student

PROFESSIONAL MEMBERSHIP

American Society of Mechanical Engineers (ASME) <i>Graduate Member: 21100218242</i>	11/2025 – Present
Nigeria Society of Engineers (NSE) <i>Corporate Member: 71046</i>	10/2025 – Present
Sigma Xi: The Scientific Research Honor Society <i>Full Member: 20259994673</i>	09/2025 – Present
American Society of Civil Engineers <i>Graduate Student Member:000012565961</i>	07/2025 – Present
Institute of Electrical and Electronics Engineers <i>Graduate Student Member:98644041</i>	07/2025 – Present
Society of Automotive Engineers (SAE) <i>Graduate Student Member:6161206268</i>	03/2025 – Present
Society for the Advancement of Material & Process Engineering (SAMPE) <i>Graduate Student Member:80071811</i>	04/2025 – Present
American Society of Composites (ASC) <i>Graduate student Member</i>	06/2024 – Present
Society of Plastic Engineers (SPE) <i>Graduate Member</i>	08/2023 – Present
YALI Regional Leadership Center <i>Cohort Member</i>	06/2020 – Present